Document 1: Retrieval-Augmented Generation (RAG) Architecture

1. Introduction
Retrieval-Augmented Generation (RAG) combines external information retrieval with text generation. Standard LLMs suffer from static knowledge and hallucination issues. By querying a vector store before generation, the model receives factual, grounded context.

2. Workflow Stages
- Ingestion: Loading raw documents, cleaning text, and splitting them into chunks.
- Embedding: Converting text chunks into vectors using embedding models.
- Retrieval: Fetching the top-k most similar chunks using cosine similarity.
- Generation: Passing the retrieved context along with the user prompt to the LLM.